11.10.6 Scope resolution operator

The scope operator reveals the hidden scope of a variable. The scope resolution operator (::) is used for the following purposes.

- To access a Global variable when there is a Local variable with same name. An example using Scope Resolution Operator.

Program 11.27

```
// Program to show that we can access a global variable
// using scope resolution operator :: when there is a local
// variable with same name //
#include<iostream>
using namespace std;
int x=45;  // Global Variable x
int main()
{
  int x = 10; // Local Variable x
  cout << "\nValue of global x is " << ::x;
  cout << "\nValue of local x is " << x;
  return 0;
}
```

Output:

```
Value of global x is 45
Value of local x is 10
```

Points to Remember:

- A large program can typically be split into smaller sized blocks called as functions.
- Functions can be classified into Pre-defined or Built-in or Library Functions and User-defined Functions.
- User-defined functions are created by the user.
- The void function tells the compiler that the function returns nothing.
- The return statement returns a value to the calling function and transfers the program control back to the calling function.
- The default return type of a function in C++ is of type int.
- A function that calls itself is known as recursive function.
- Scope refers to the accessibility of a variable.
- There are four types of Scopes. They are: Local scope, Function scope, File scope and Class scope.
- The scope operator (::) reveals the hidden scope of a variable.

11th Std Computer Science_EM Chapter 11.indd 203 06-01-2023 15:10:20